

EXERCISE

PROBLEM: A key step in the production of sulfuric acid is the oxidation of $\text{SO}_2(\text{g})$ to $\text{SO}_3(\text{g})$:



At 298 K, $\Delta G = -141.6 \text{ kJ/mol}$; $\Delta H = -198.4 \text{ kJ/mol}$; and $\Delta S = -187.9 \text{ J/K}$

- (a) Use the data to decide if this reaction is spontaneous at 25 deg. C, and predict how ΔG will change with increasing T .
- (b) Assuming ΔH and ΔS are constant with increasing T , is the reaction spontaneous at 900 deg.C?

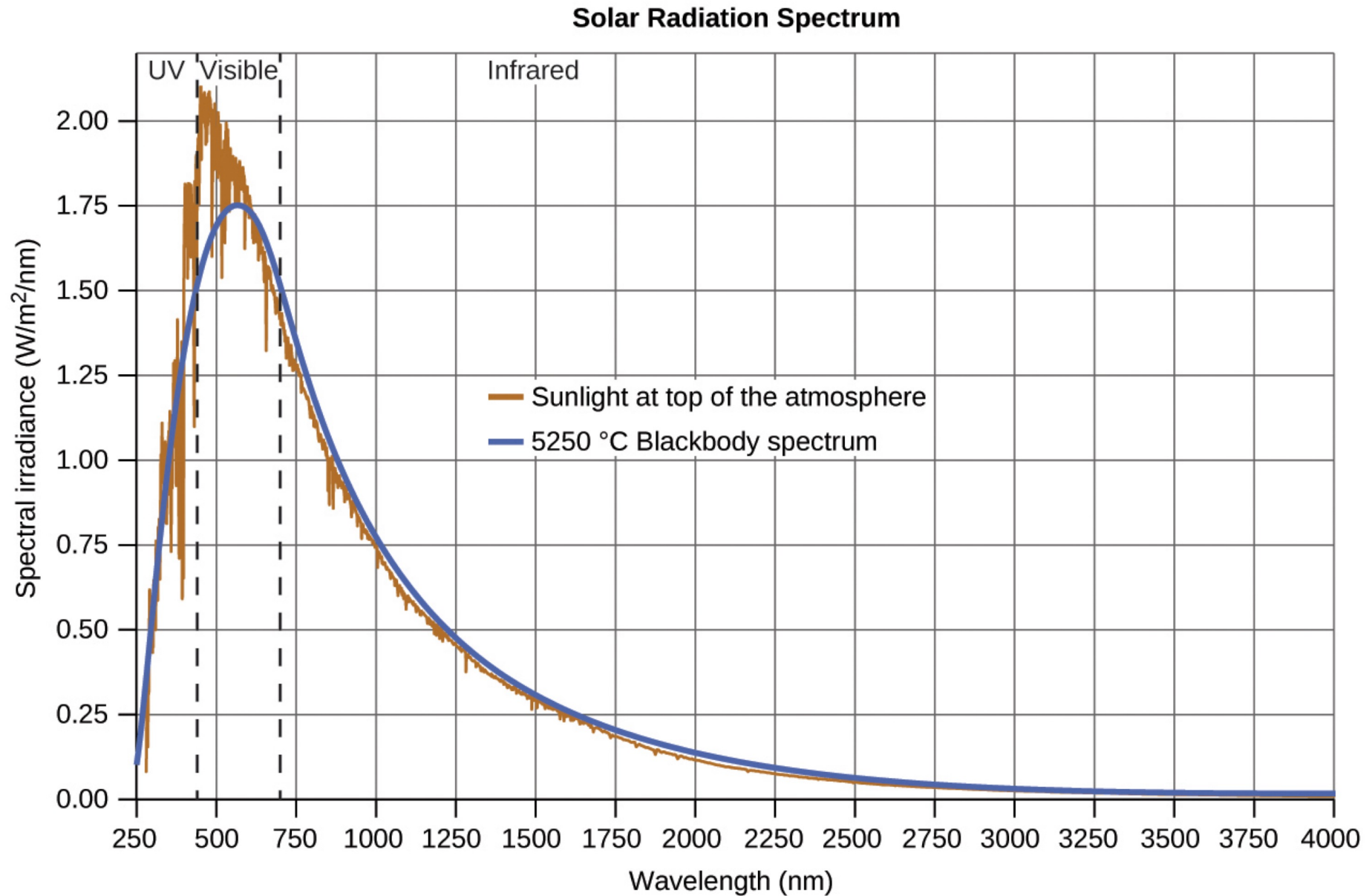
The Electronic Structure of Atoms & The Periodic Table

26021 Chemistry (Polytechnical Foundation)

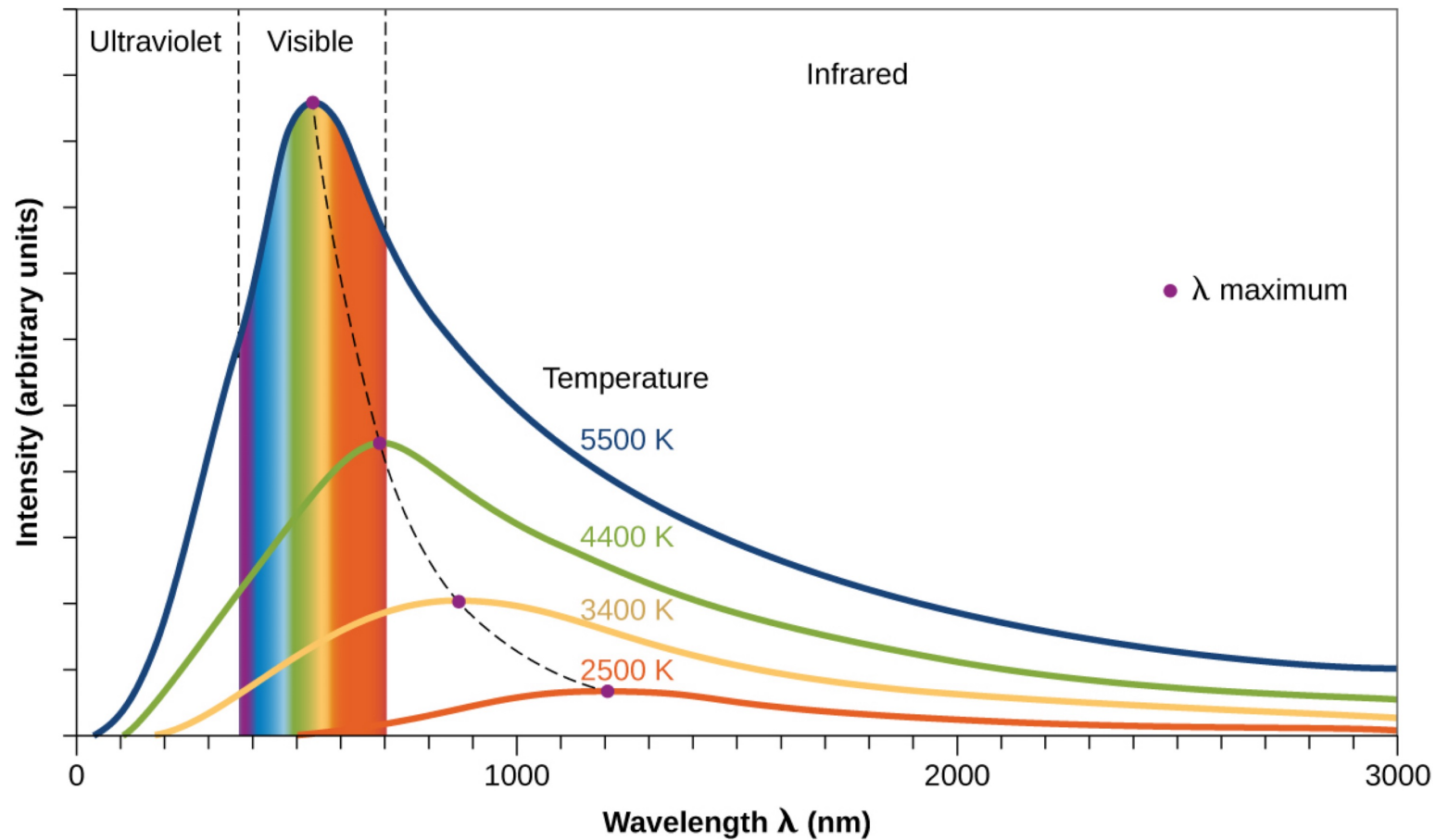
Professor Kasper S. Pedersen

2026

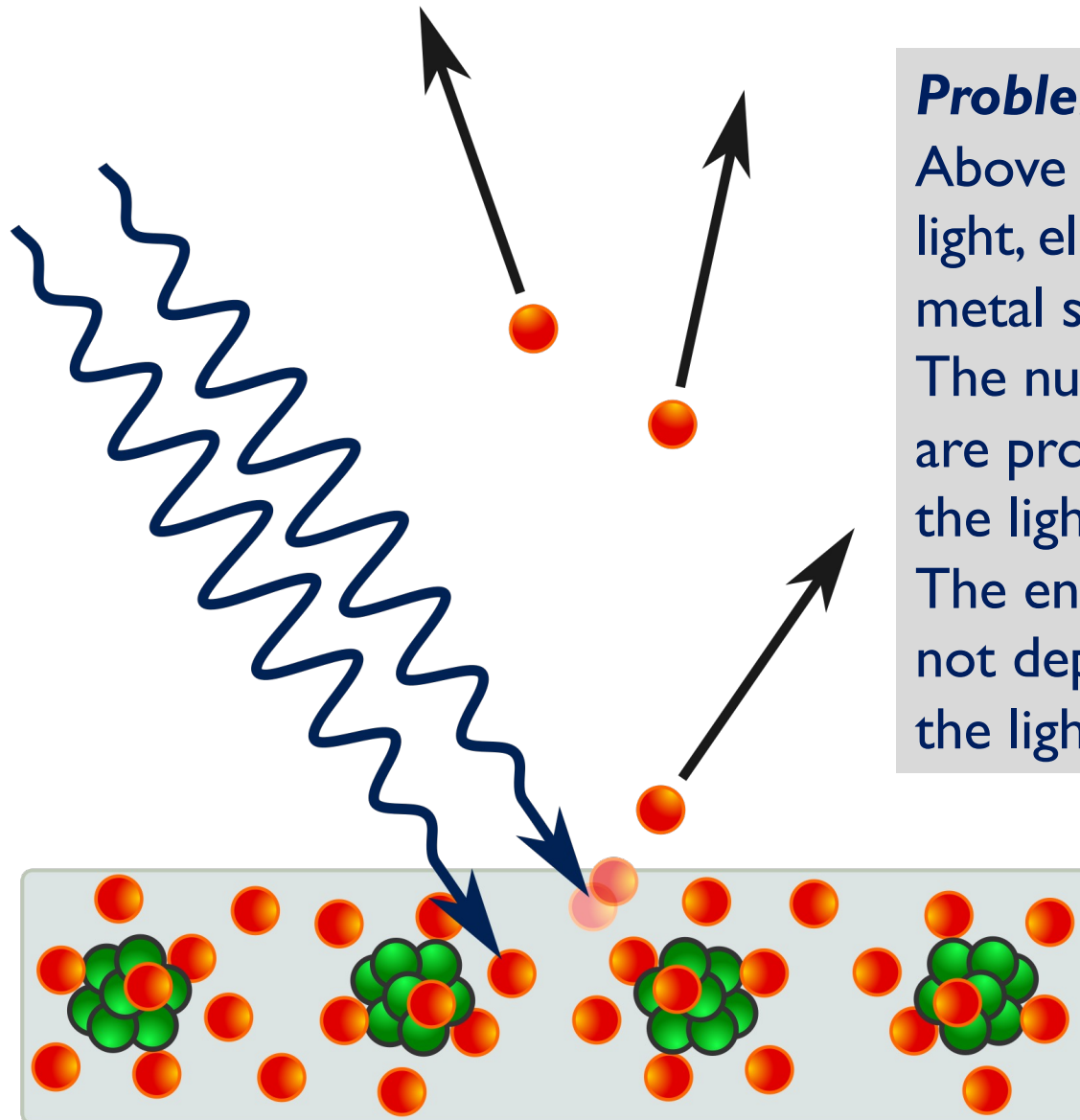
Black-body radiation



Black-body radiation



Photoelectric effect

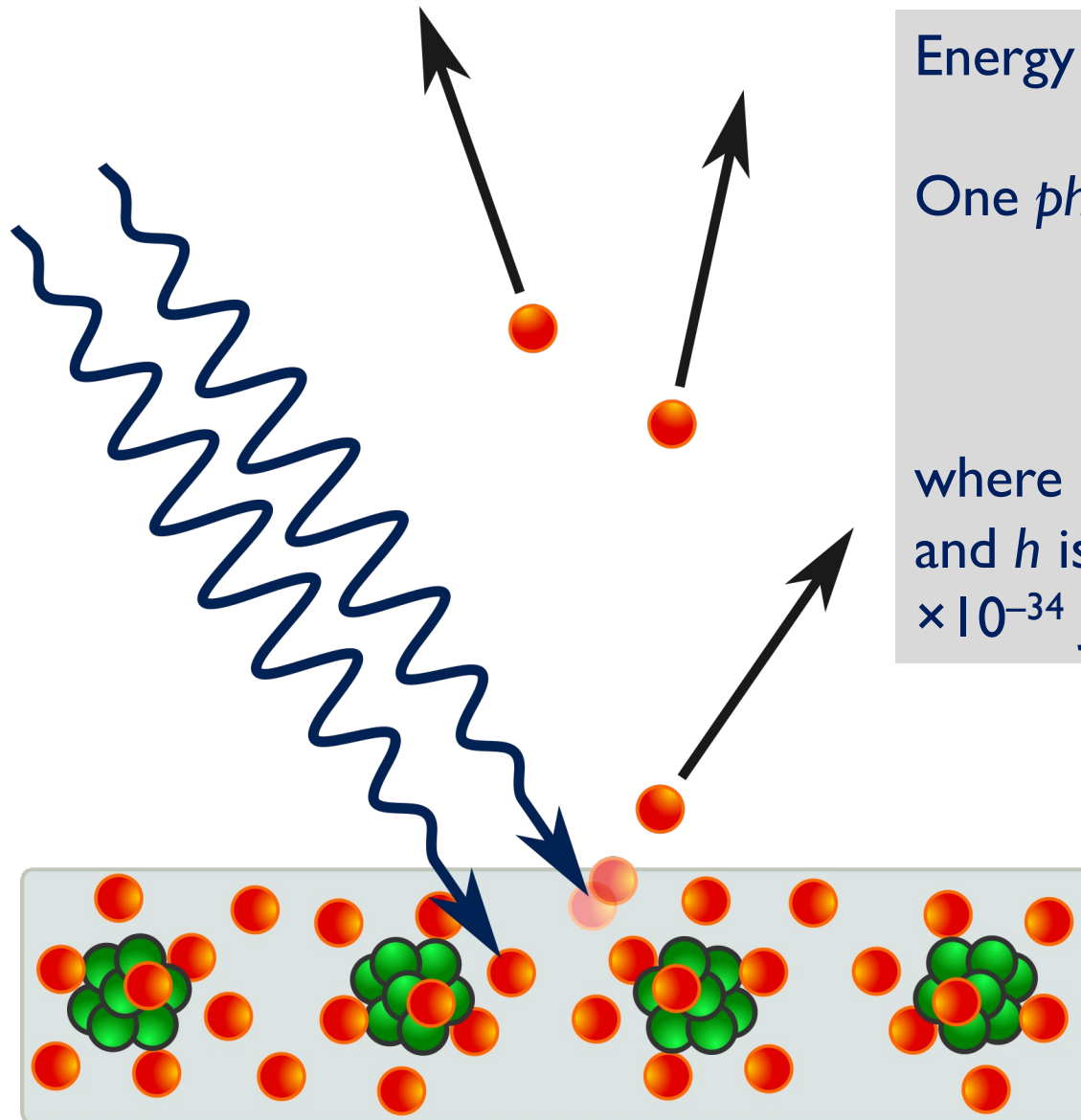
***Problem:***

Above a threshold frequency of light, electrons are expelled from a metal surface.

The number of expelled electrons are proportional to the intensity of the light.

The energies of the electrons were not dependent on the intensity of the light!

Photoelectric effect

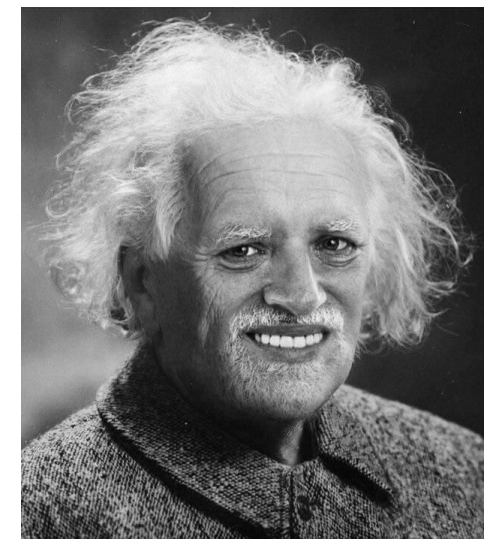


Energy is quantized (Planck, 1900).

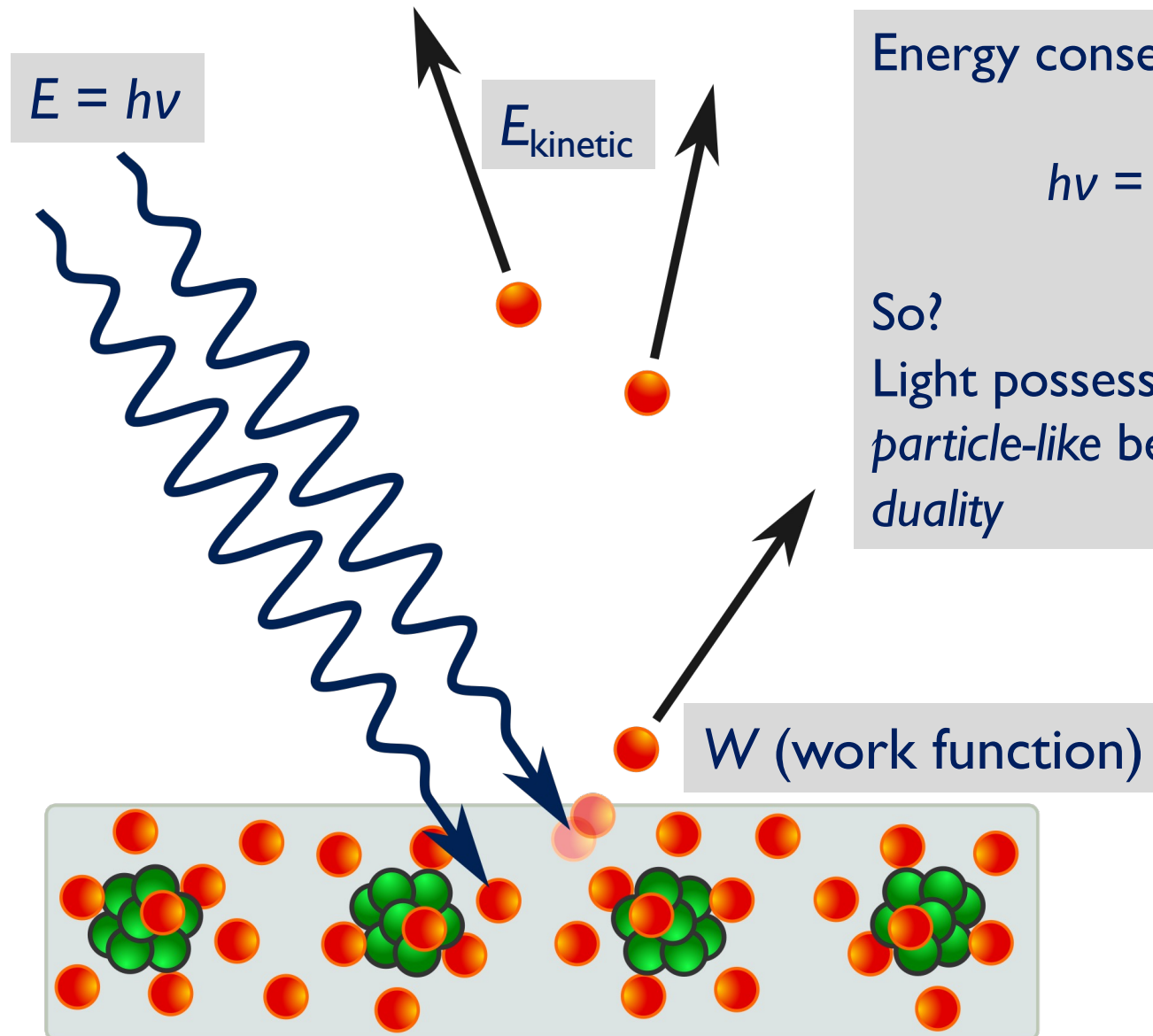
One *photon* has an energy of

$$E = h\nu$$

where ν is the *frequency* of the light and h is the Planck constant ($6.63 \times 10^{-34} \text{ J s}$)



Photoelectric effect

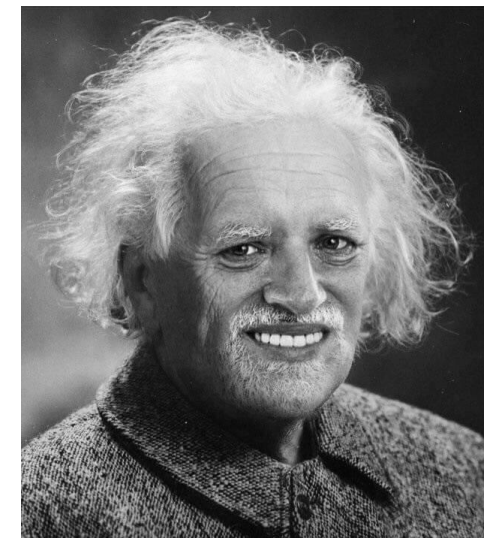


Energy conservation means

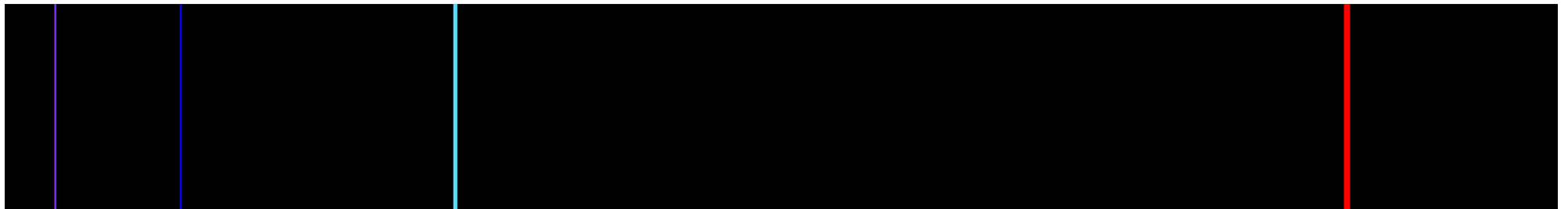
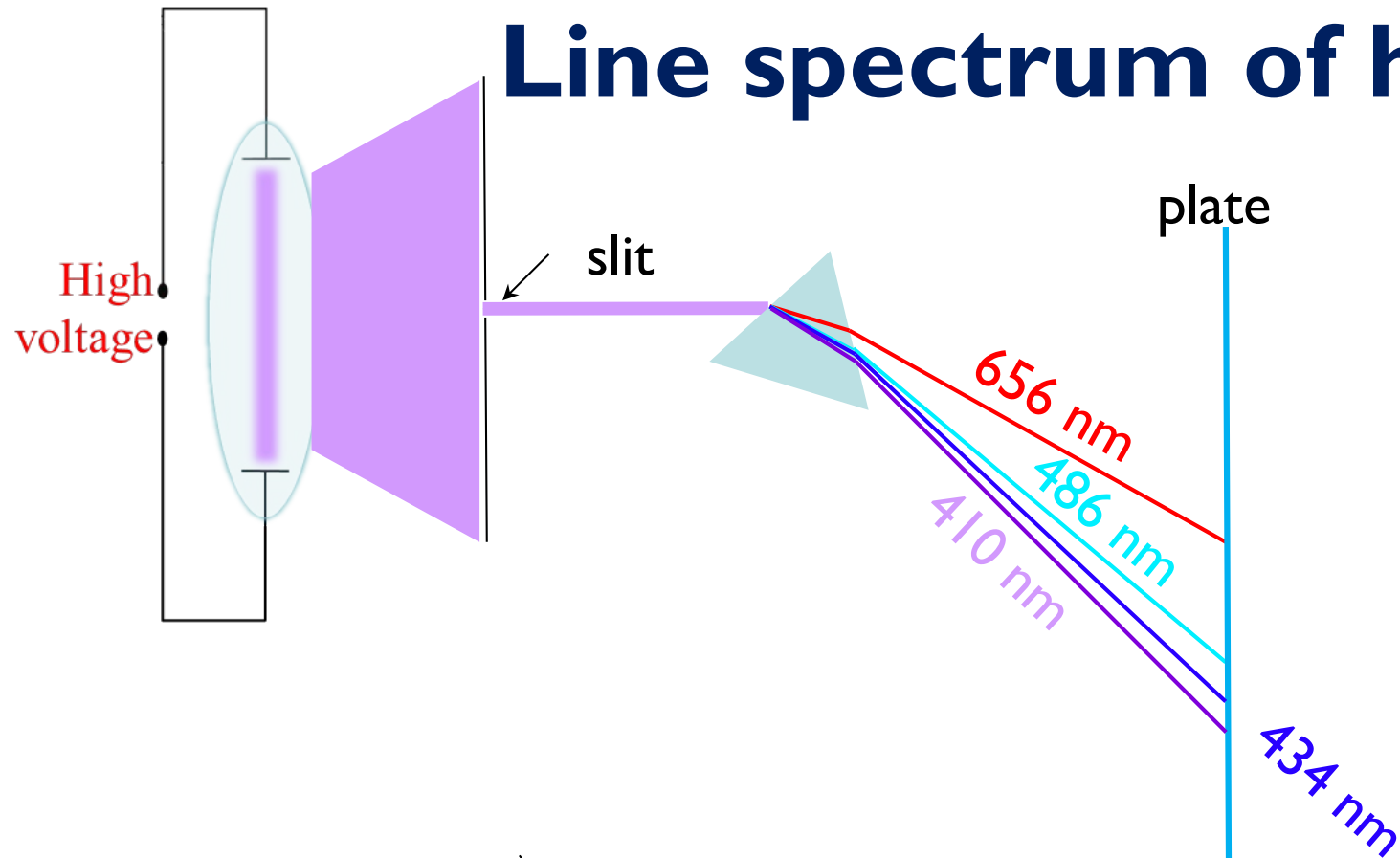
$$h\nu = E_{\text{kinetic}} + W$$

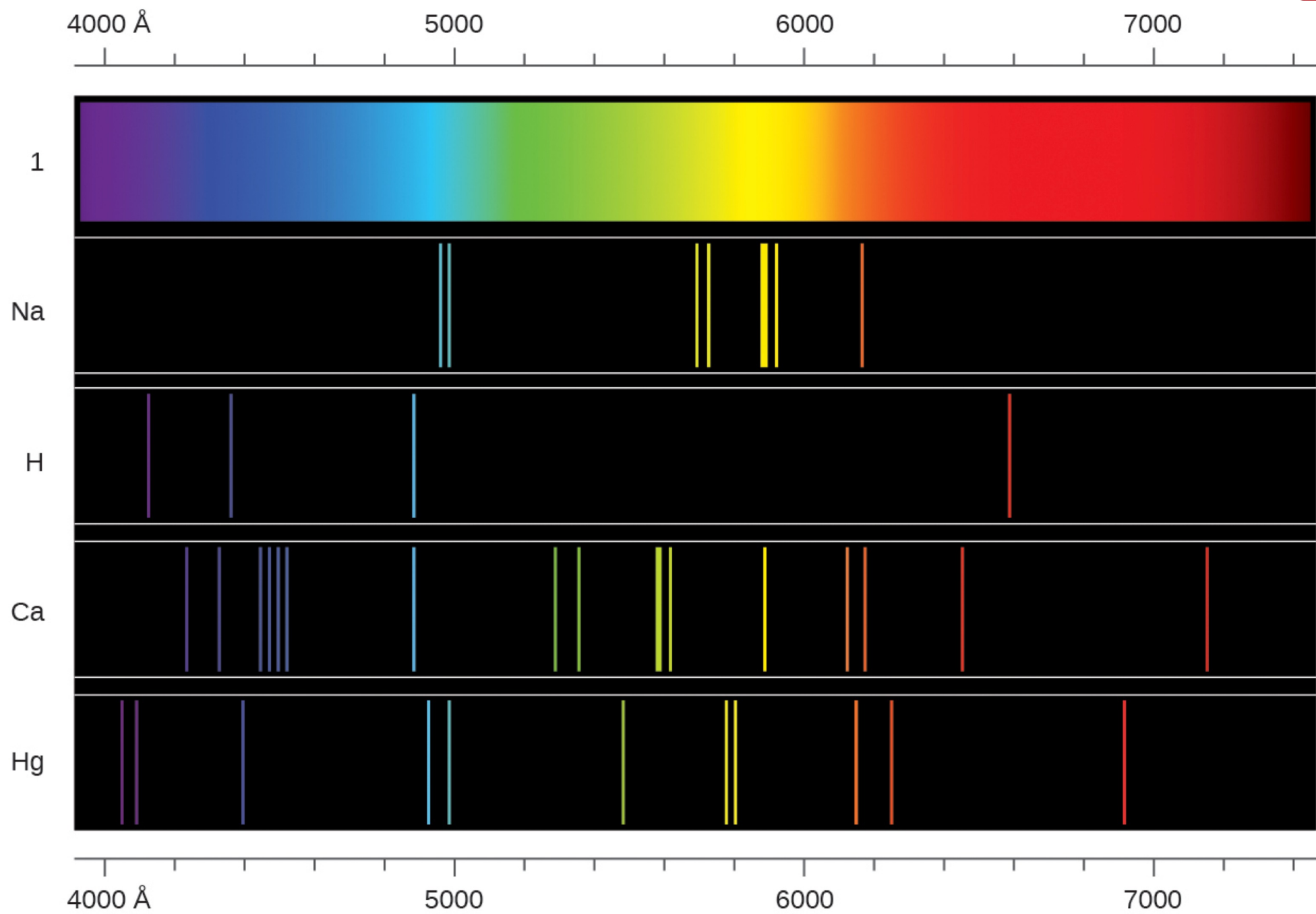
So?

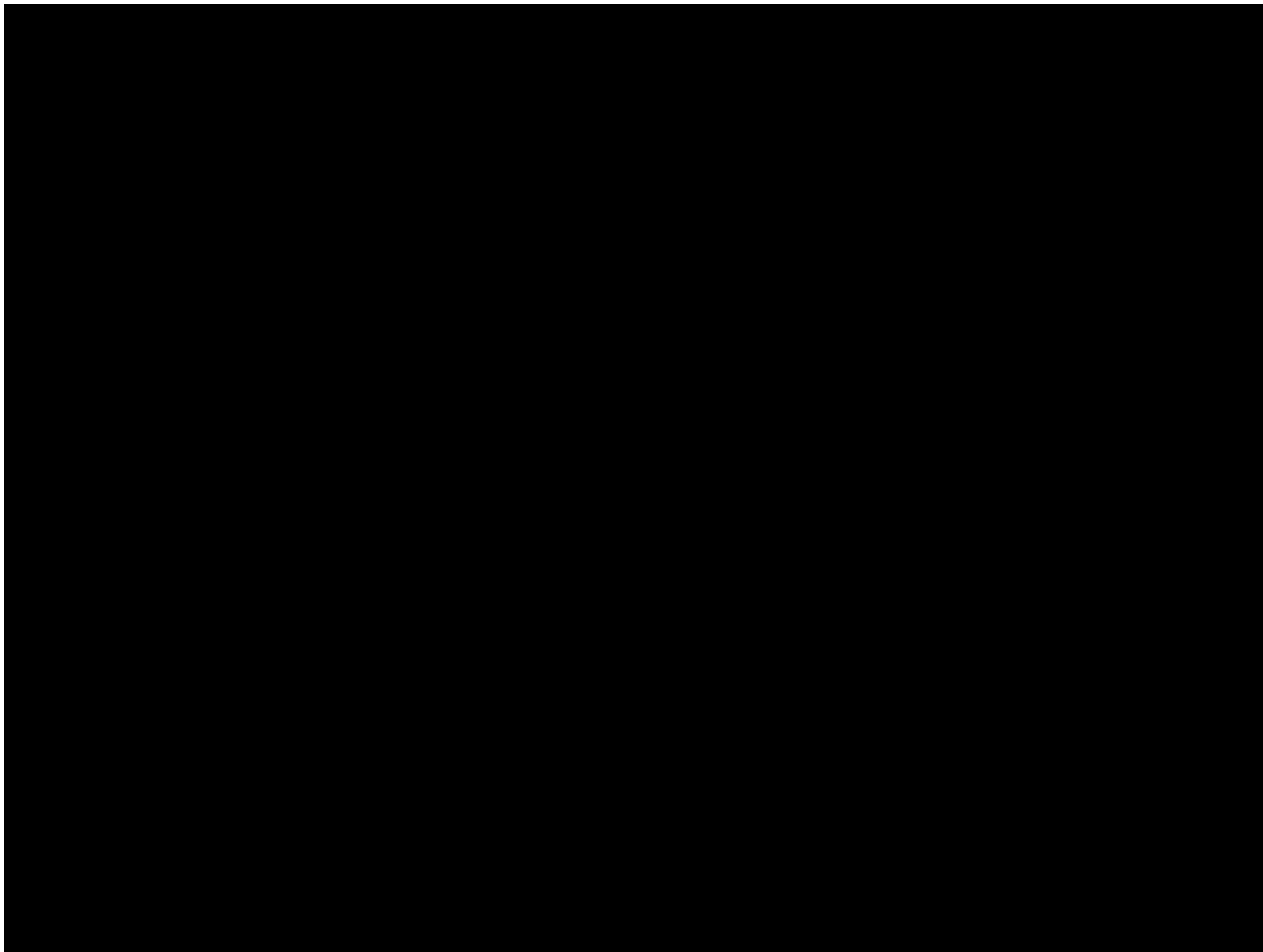
Light possesses both *wave-like* and *particle-like* behaviour: *particle-wave duality*



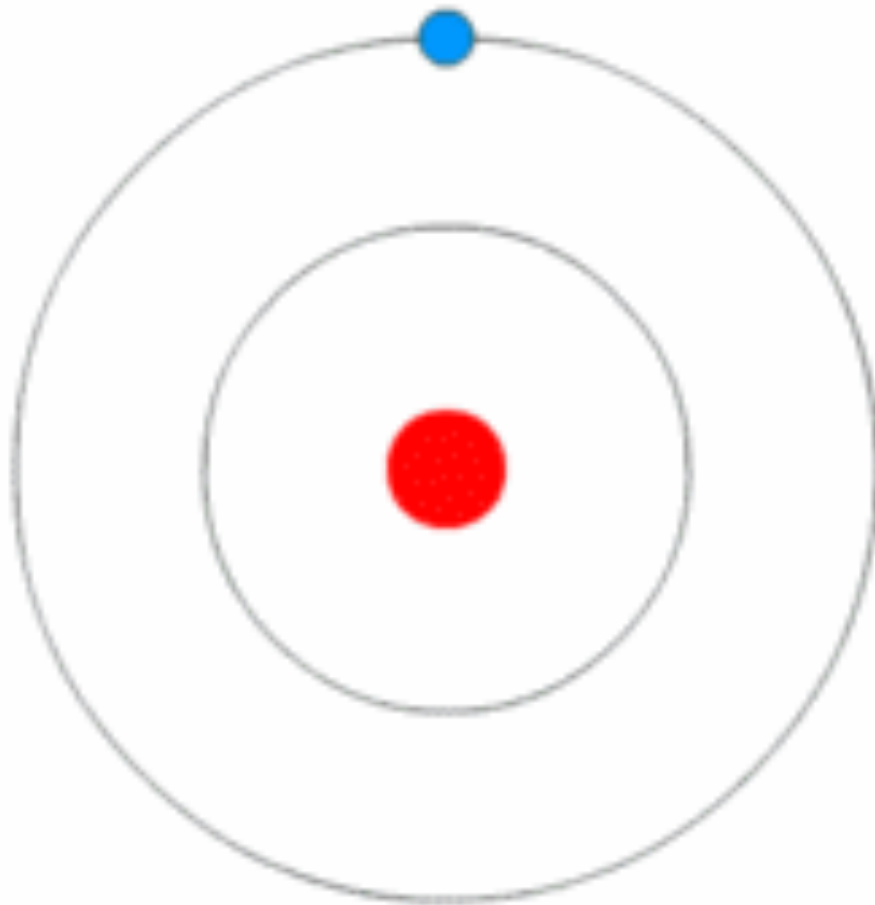
Line spectrum of hydrogen







Niels Bohr's Model



1. Electrons in an atom can only occupy certain orbits.

2. Electrons in permitted orbits have specific, “allowed” energies, given by

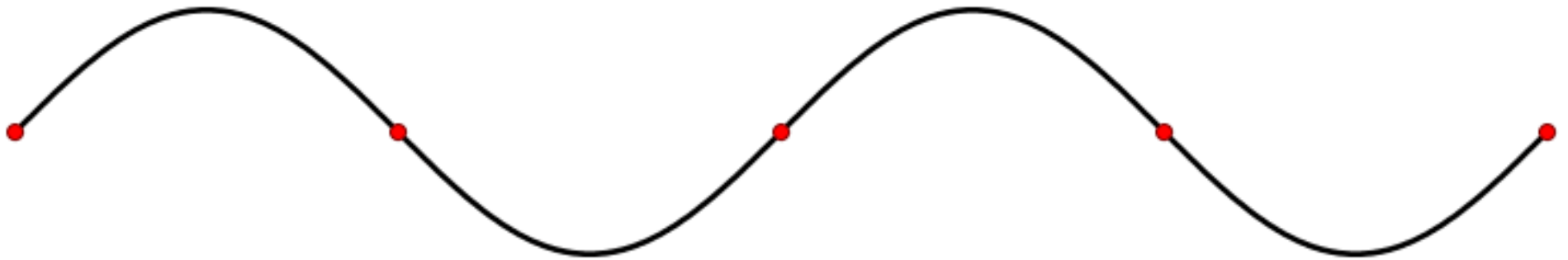
$$E_n = -R_H n^{-2}$$

where n is the *principal quantum number*

3. Energy is only absorbed or emitted in such a way as to move an electron from one “allowed” energy state to another.

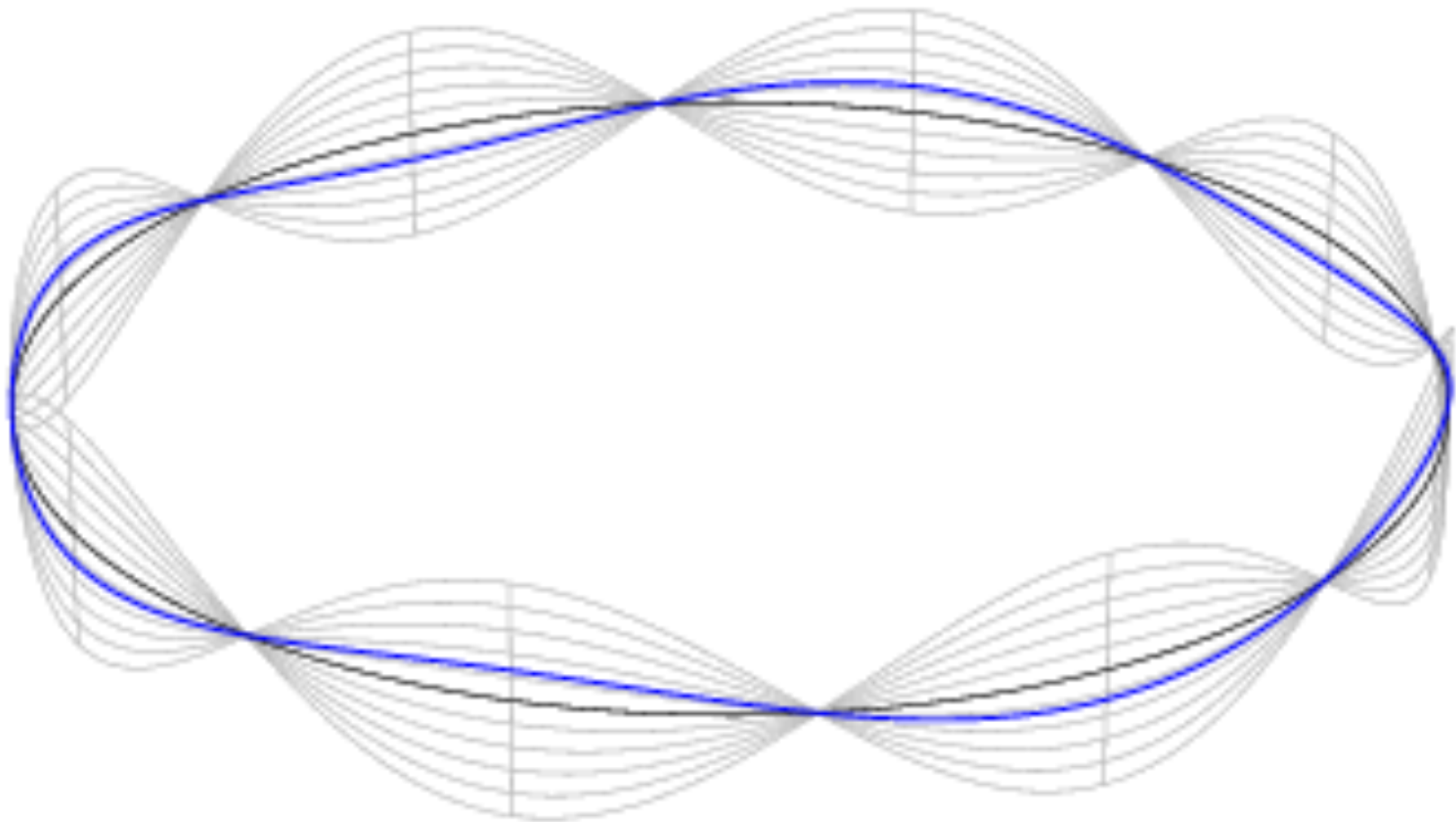
Electrons are standing waves?

De Broglie's idea



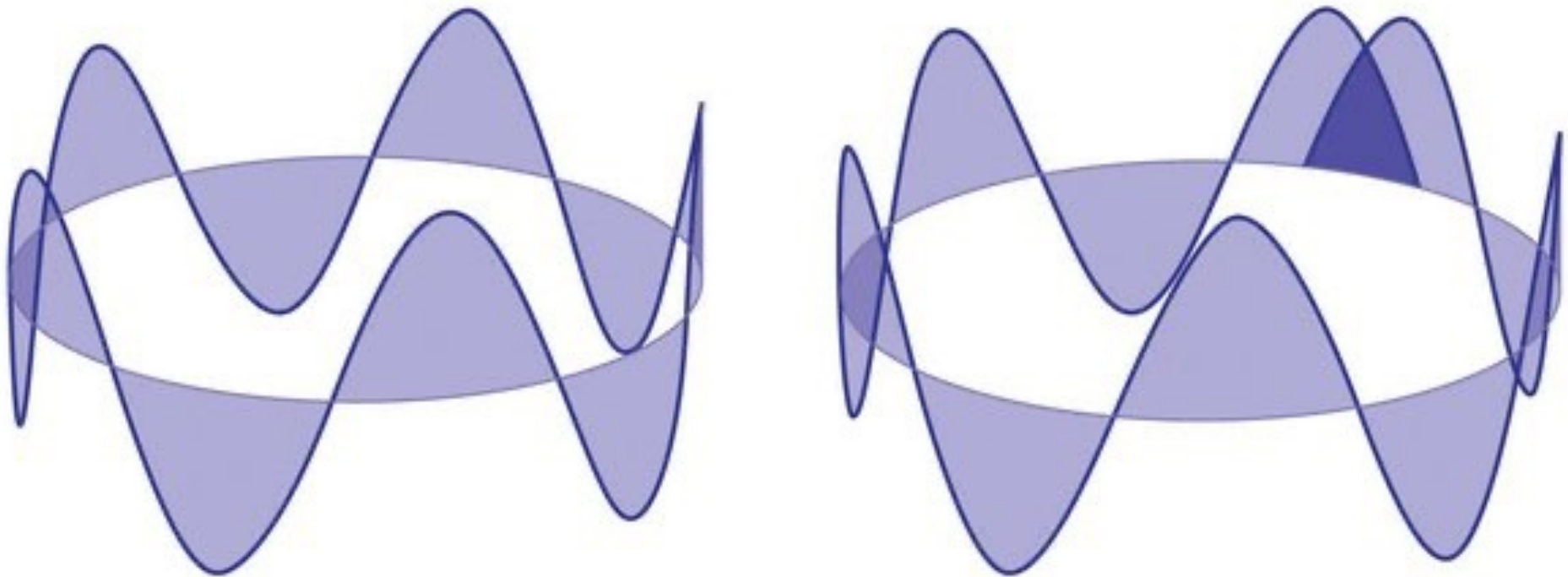
Electrons are standing waves?

De Broglie's idea



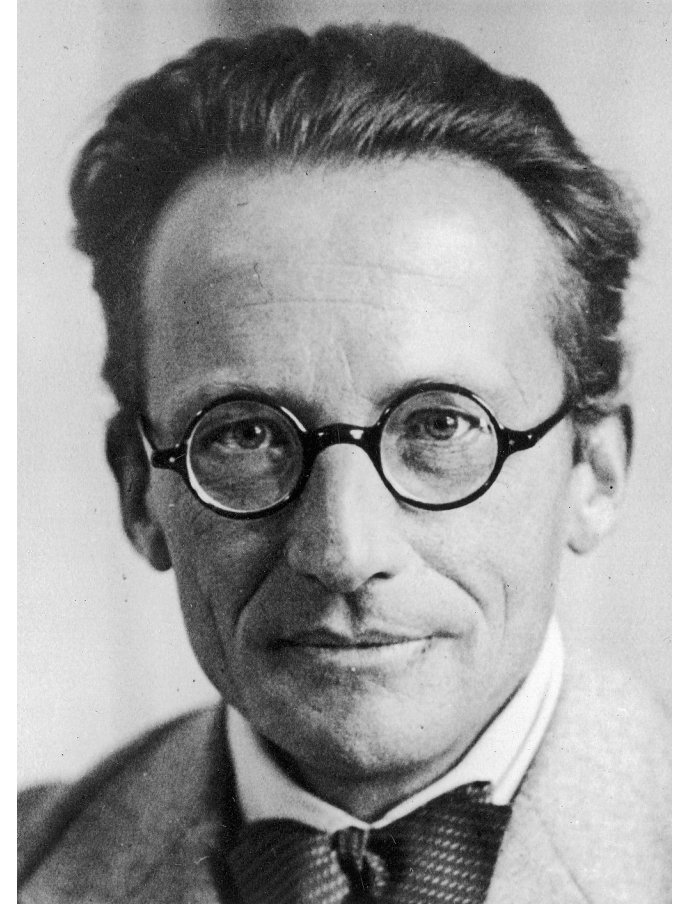
Electrons are standing waves?

De Broglie's idea



Schrödinger's Equation 🤯

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$



Quantum numbers

Quantum numbers and atomic orbitals

n	ℓ	m_ℓ	# of orbitals	atomic orbital designation
1	0	0	1	1s
2	0	0	1	2s
	1	-1, 0, 1	3	$2p_x, 2p_y, 2p_z$
3	0	0	1	3s
	1	-1, 0, 1	3	$3p_x, 3p_y, 3p_z$
	2	-2, -1, 0, 1, 2	5	$3d_{xy}, 3d_{yz}, 3d_{yz}, 3d_{x^2-y^2}, 3d_{z^2}$
.	.	.	.	.

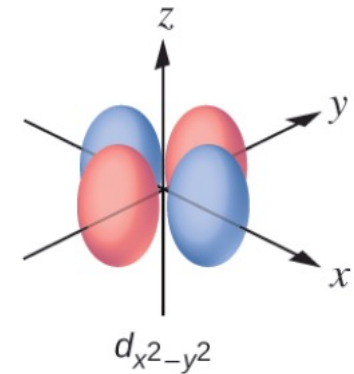
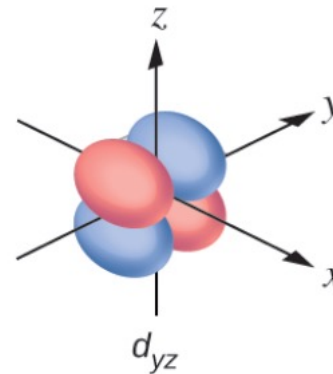
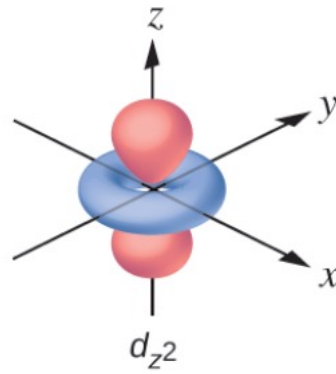
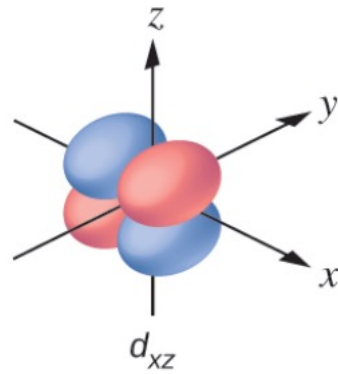
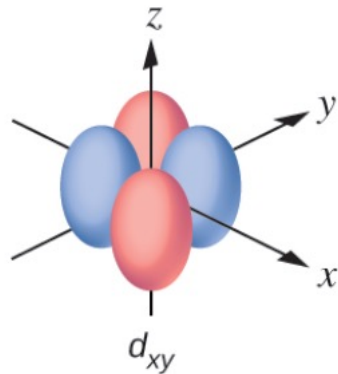
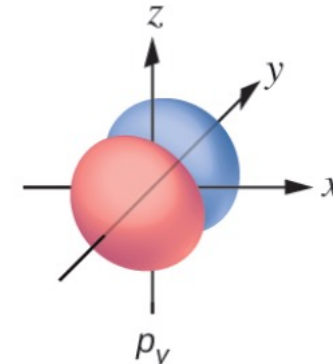
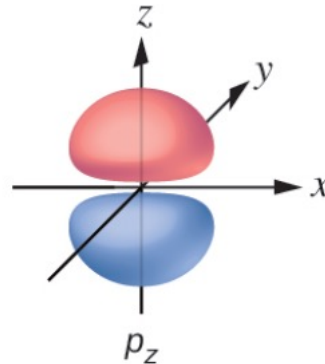
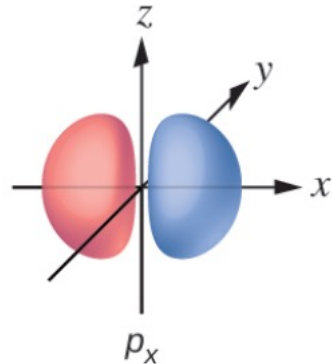
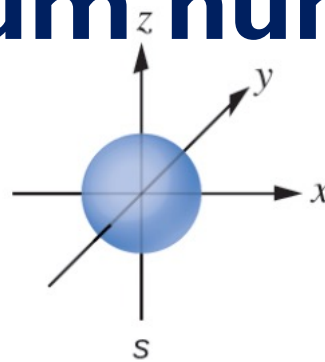
$n = 1, 2, 3, 4...$
main numbers:
shell

$\ell = 0, 1, 2, 3... n-1$
angular momentum
numbers
shape:
s, p, d, f
orbitals

$m = -\ell, -\ell+1...0... \ell-1, \ell$
magnetic numbers:
direction
position
of orbitals

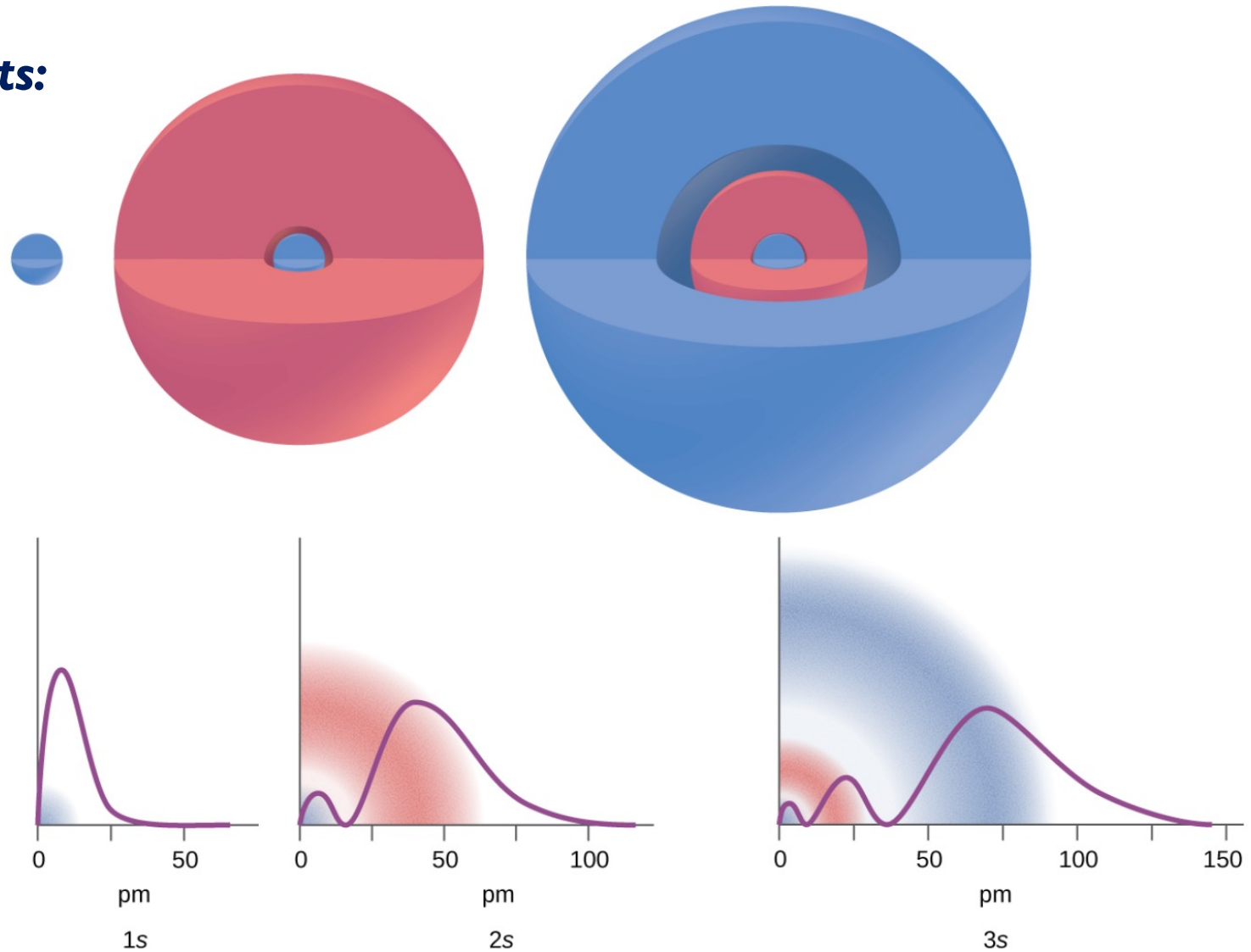
A set of quantum numbers define an orbital

Angular parts:

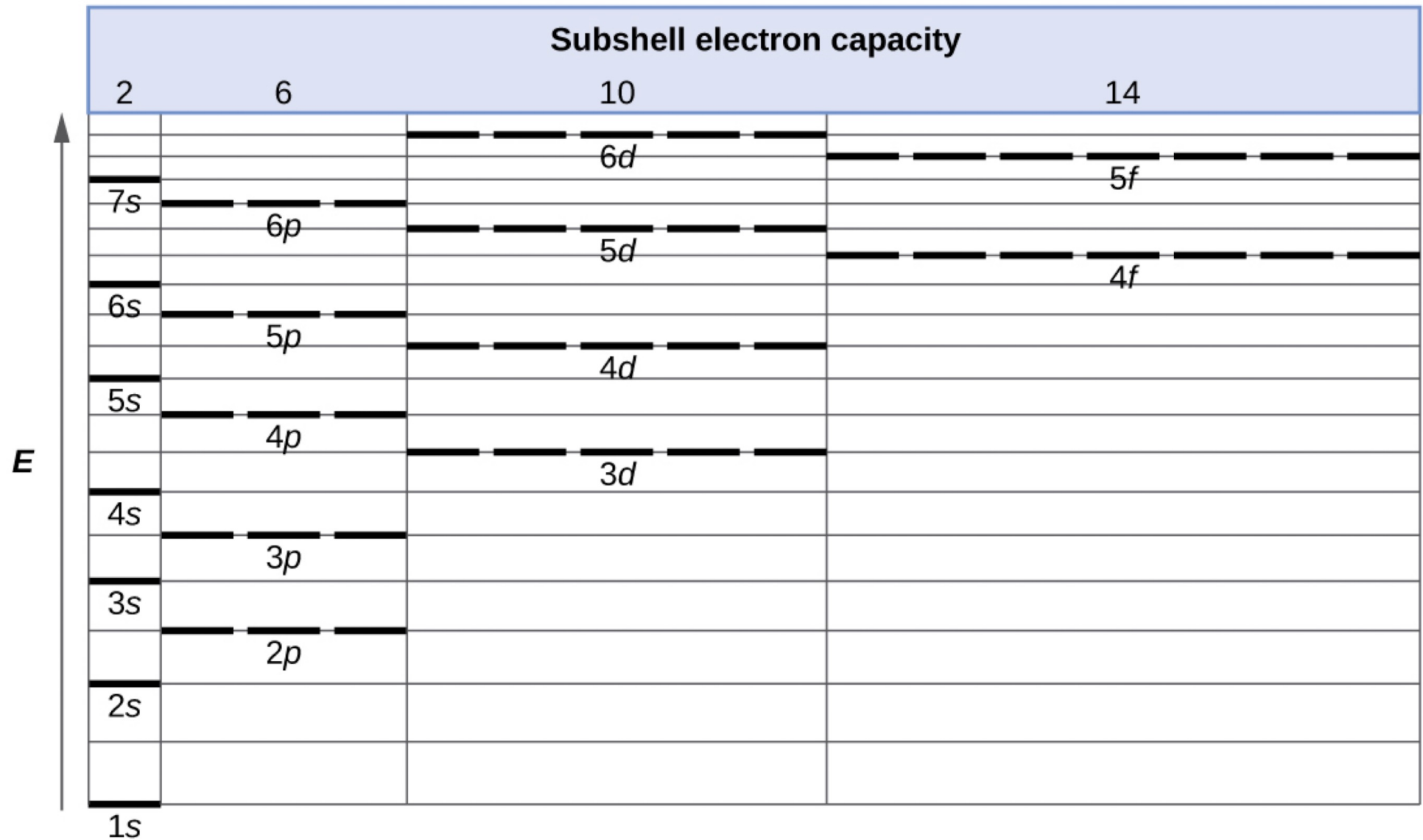


A set of quantum numbers define an *orbital*

Radial parts:

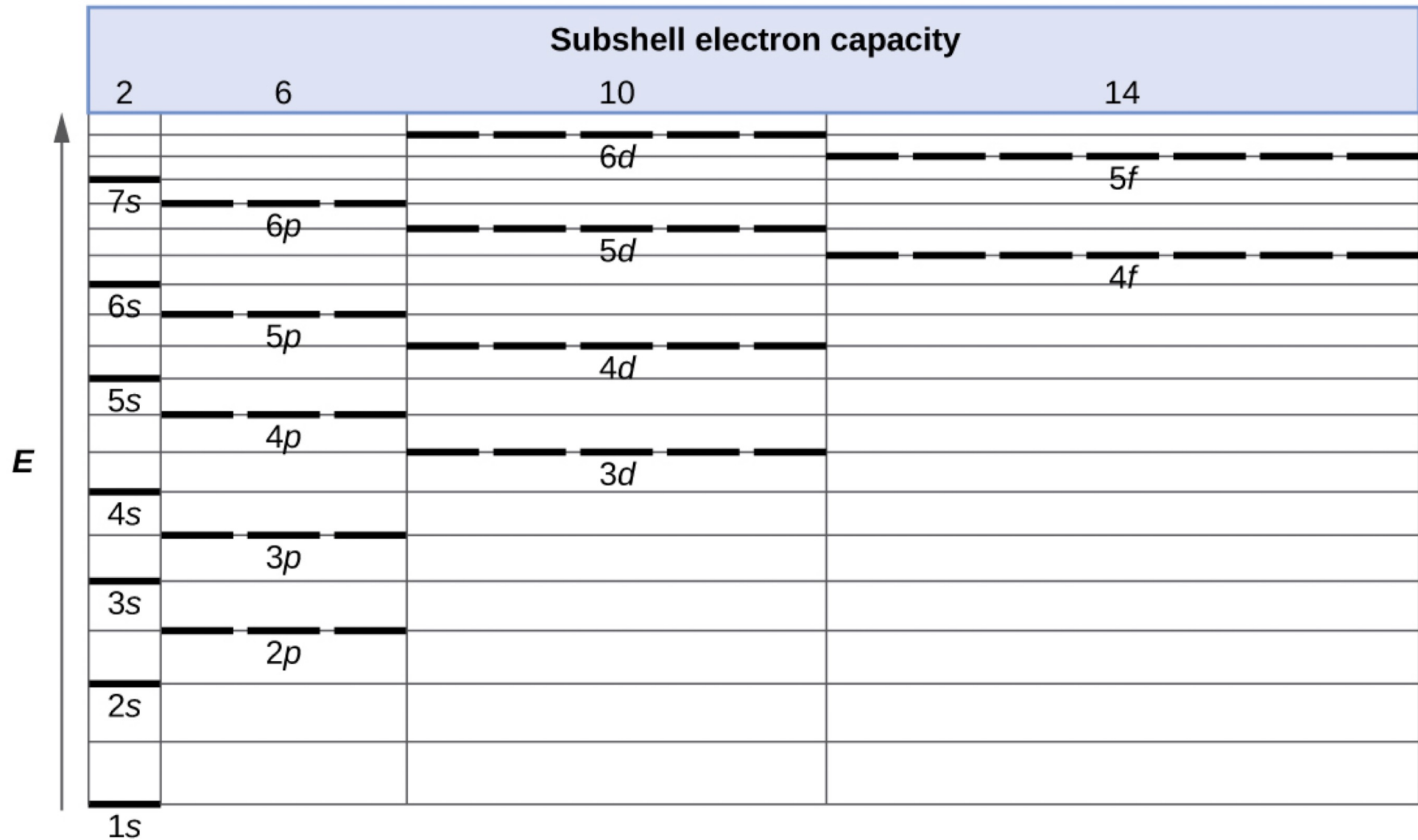


Orbital Energies & Aufbau Principle

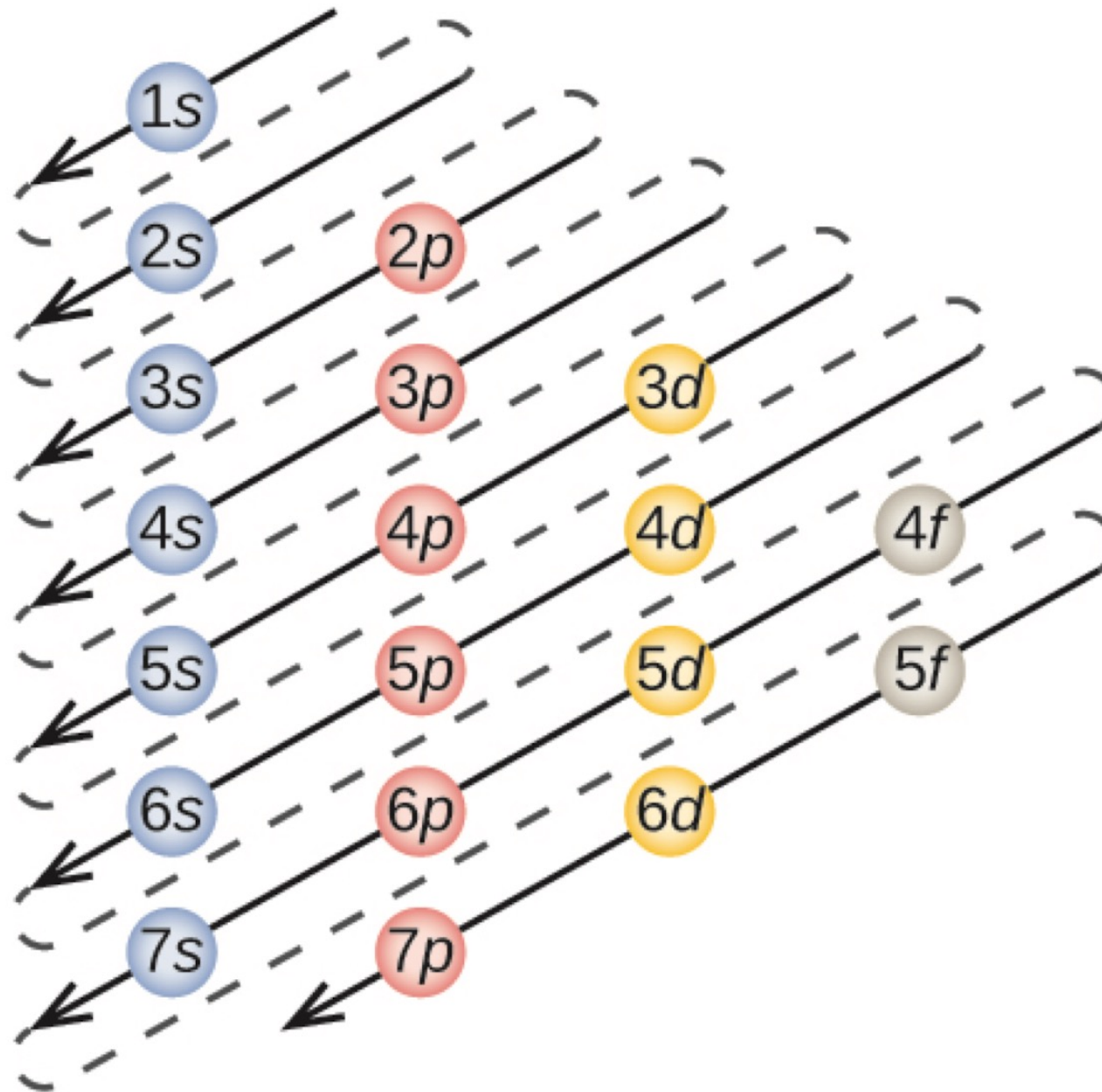


	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

Orbital Energies & Aufbau Principle

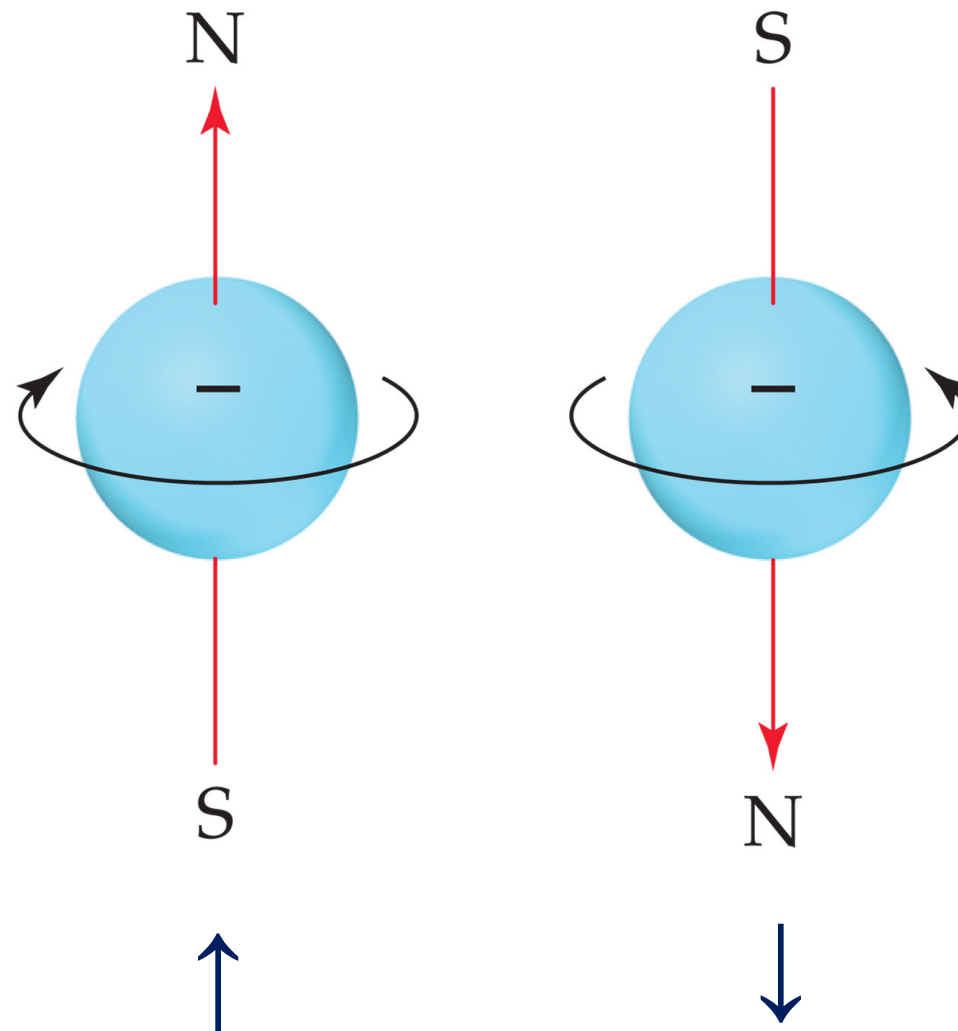


Orbital Energies & Aufbau Principle



Spin Quantum Number, m_s

Electron spin (conceptualized as rotation, in reality, no rotation...)



Atom and ion electronic structure

How to put electrons in orbitals??

- ▶ Fill the lowest energy orbitals first

Energy increases with the sum $n + l$:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, ...

- ▶ The set of quantum numbers for an electron is unique, i.e., two electrons cannot have the same four quantum numbers.

Pauli exclusion principle

- ▶ For a given n, l quantum number the most stable state maximizes $m_{s,total} = \sum_i m_{s,i}$

Hund's rule

Example: The Oxygen Atom

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

Example: The Oxygen Atom

CORRECT



WRONG
(Hund's rule)



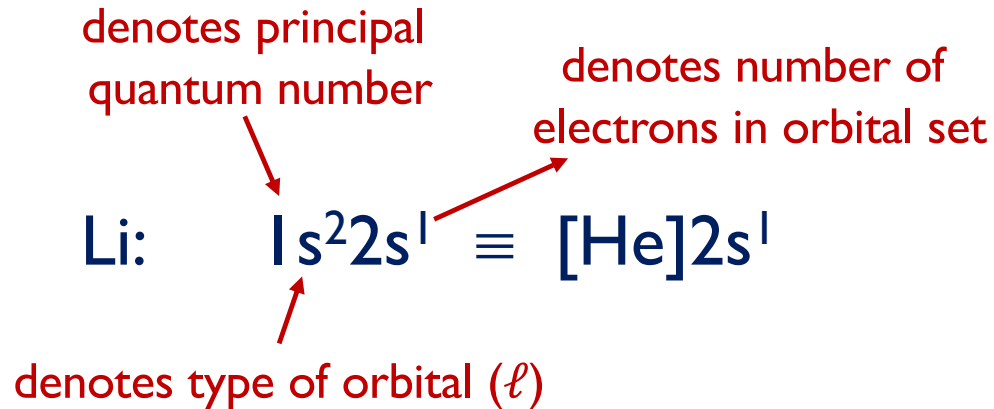
WRONG
(Hund's rule)



WRONG!!
(Pauli exclusion principle)



Electron configuration



Atomic number	Atom symbol	Electron configuration
1	H	$1s^1$
2	He	$1s^2$
3	Li	$[\text{He}] 2s^1$
4	Be	$[\text{He}] 2s^2$
5	B	$[\text{He}] 2s^2 2p^1$
6	C	$[\text{He}] 2s^2 2p^2$
7	N	$[\text{He}] 2s^2 2p^3$
8	O	$[\text{He}] 2s^2 2p^4$
9	F	$[\text{He}] 2s^2 2p^5$
10	Ne	$[\text{He}] 2s^2 2p^6$
11	Na	$[\text{Ne}] 3s^1$
12	Mg	$[\text{Ne}] 3s^2$
13	Al	$[\text{Ne}] 3s^1 3p^1$

Exercise I

What is the electronic configuration for:

(a) Na, (b) Na⁺, (c) P, (d) P³⁻?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

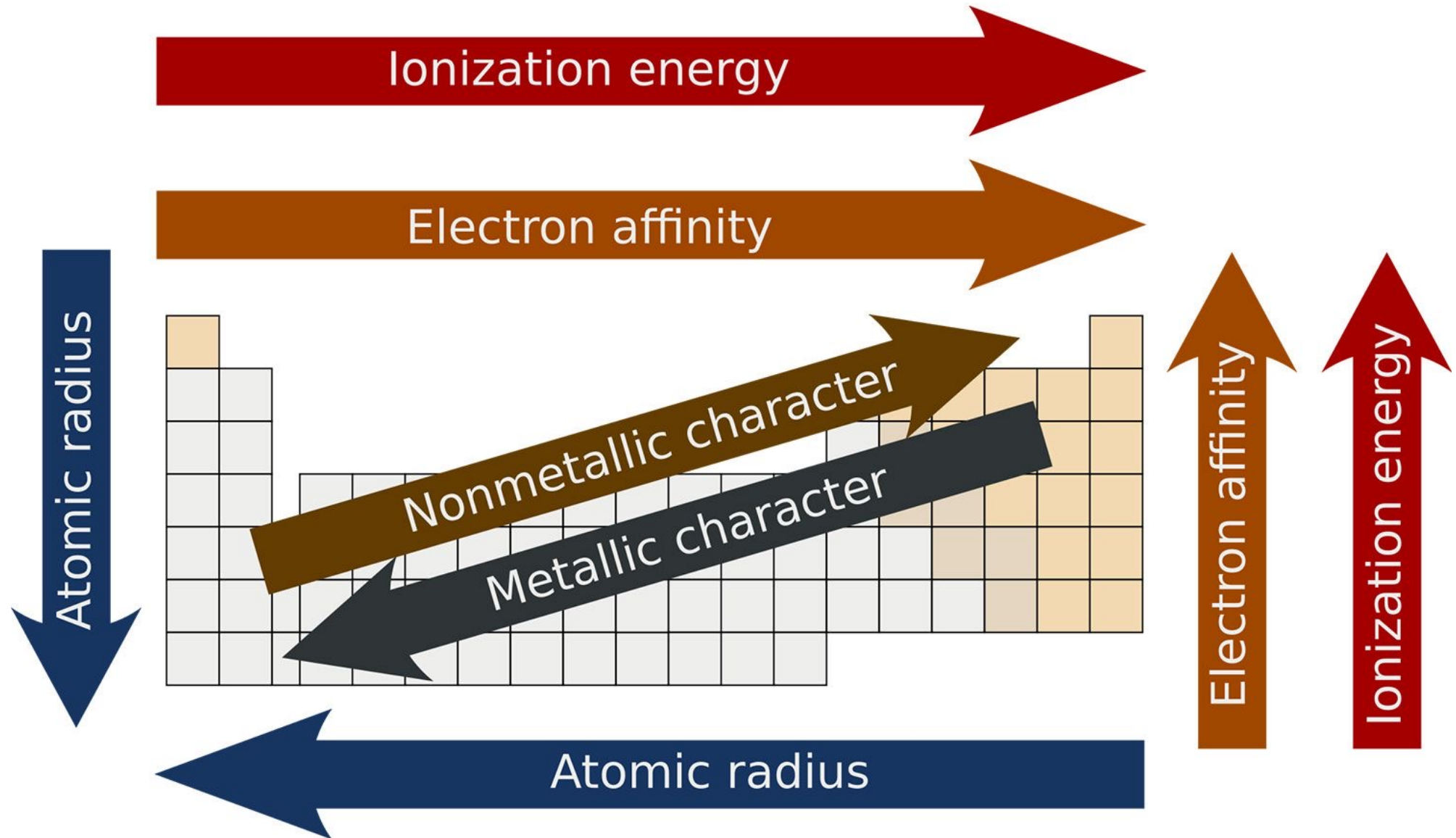
Exercise 2

What is the electronic configuration for:

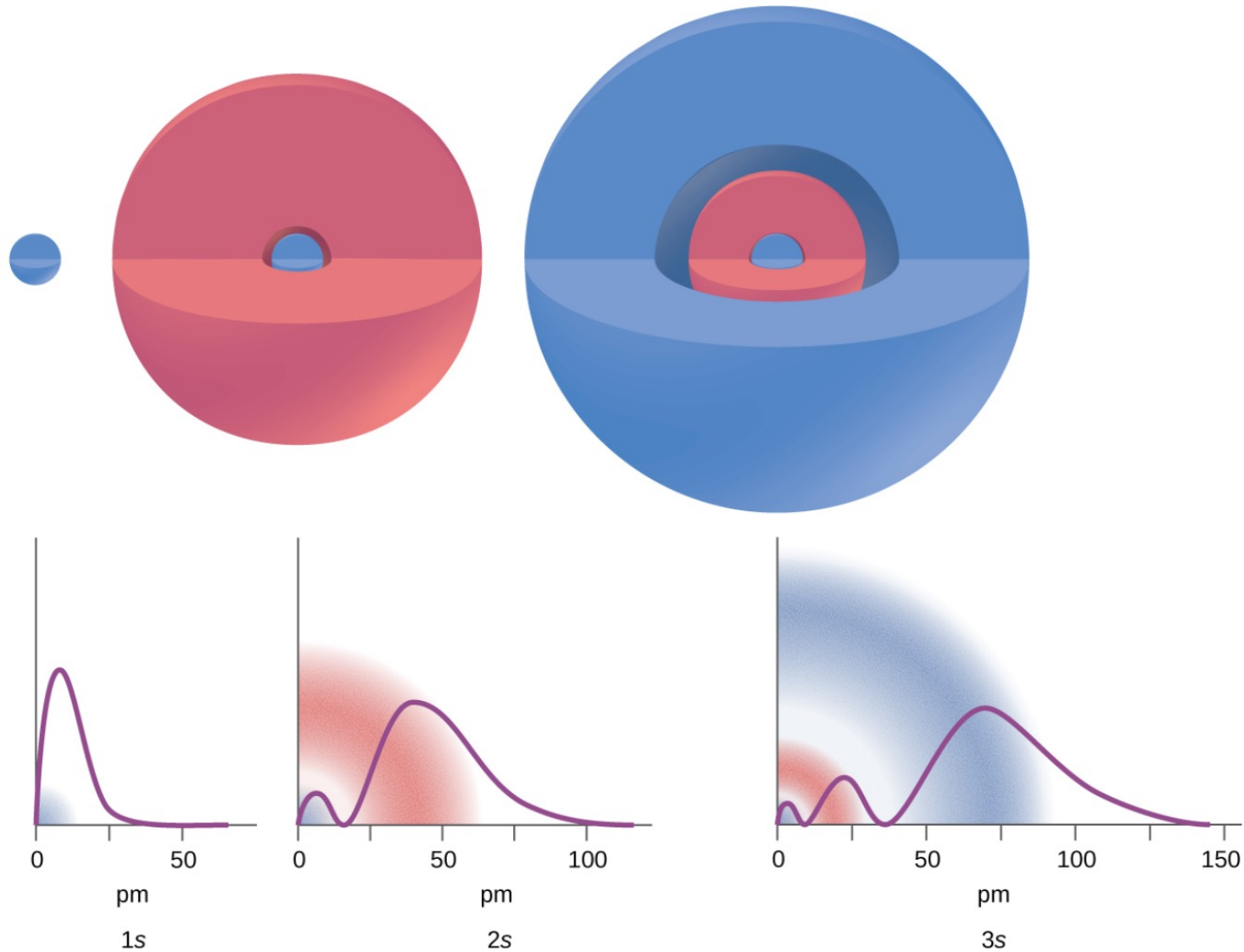
(a) Sc, (b) Se^{2-} ?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

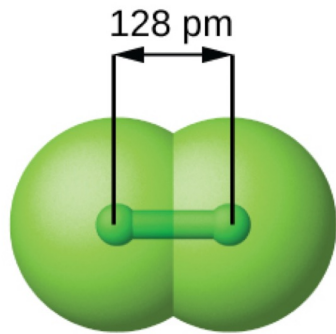
Periodic Trends in the Periodic Table



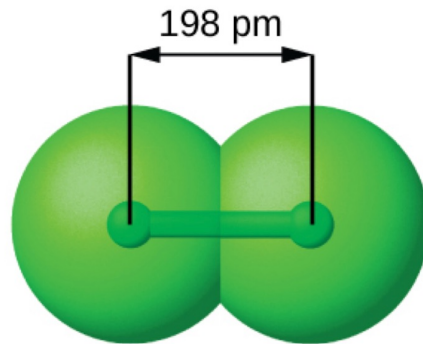
How big is an atom?



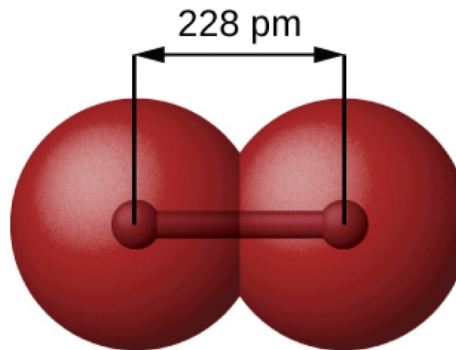
How big is an atom?



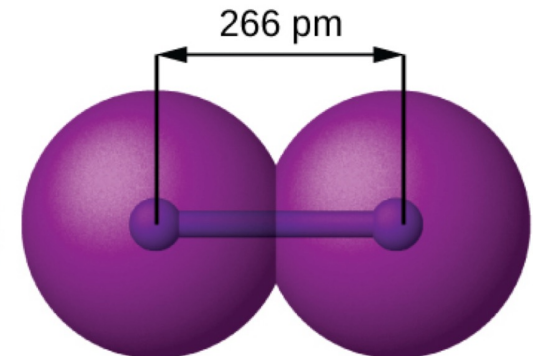
$$\text{F radius} = \frac{128 \text{ pm}}{2} = 64 \text{ pm}$$



$$\text{Cl radius} = \frac{198 \text{ pm}}{2} = 99 \text{ pm}$$



$$\text{Br radius} = \frac{228 \text{ pm}}{2} = 114 \text{ pm}$$



$$\text{I radius} = \frac{266 \text{ pm}}{2} = 133 \text{ pm}$$

Effective nuclear charge (Z_{eff})

Across a period:

	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
Z_{eff}	1.3	1.9	2.4	3.1	3.8	4.5	5.1	5.8

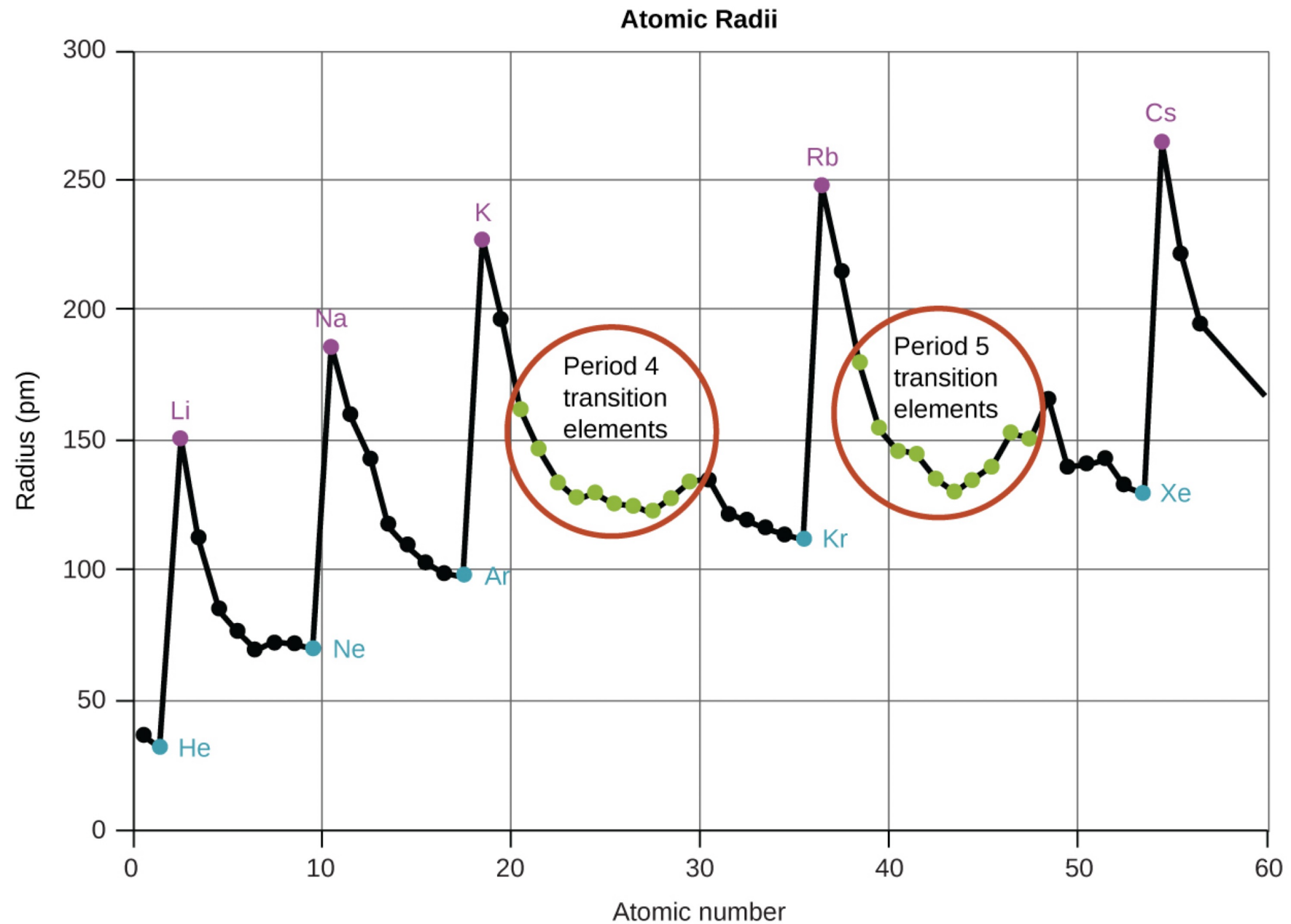
Atomic radius *decreases* along the period

Down a group:

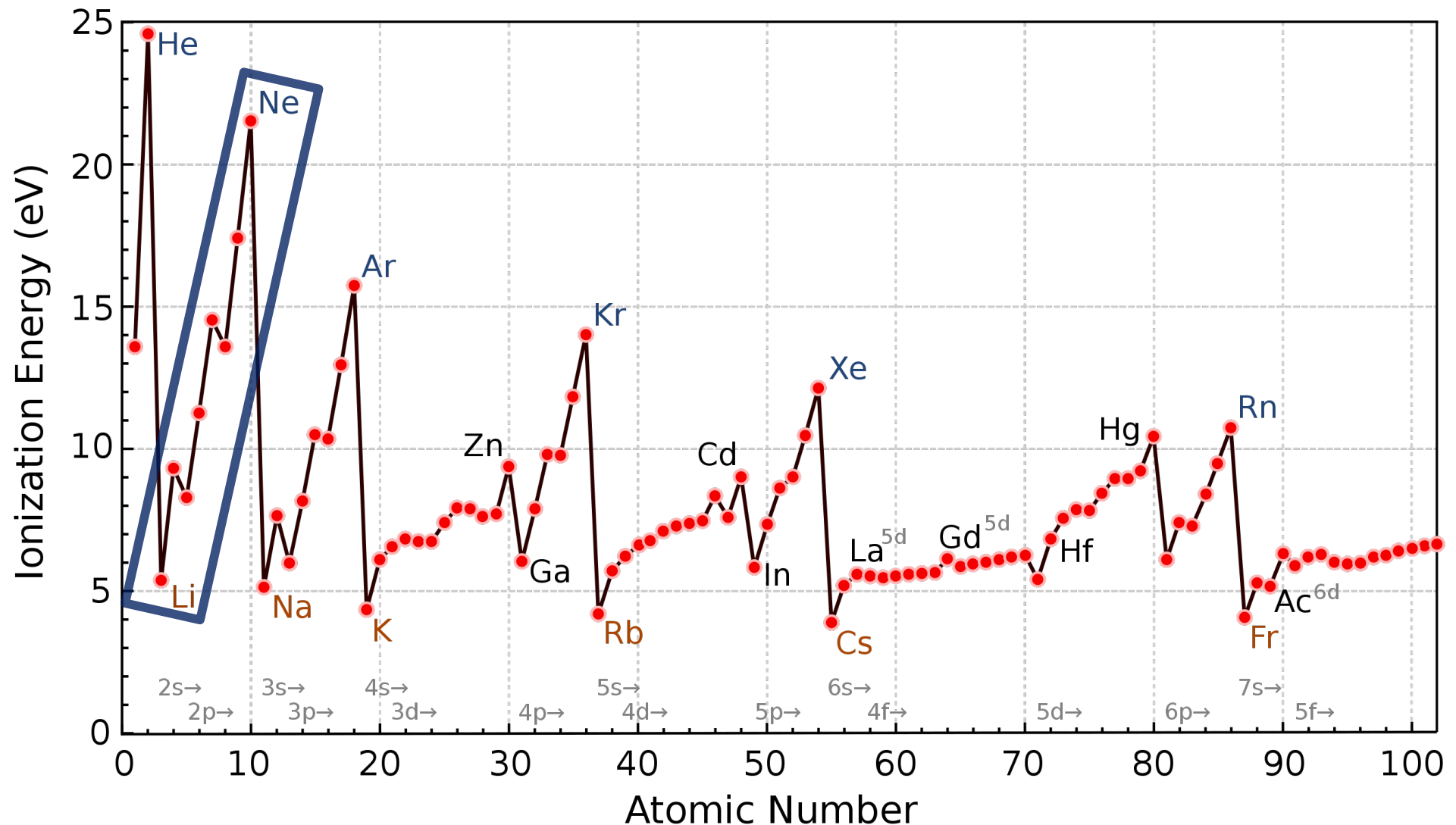
	Li	Na	K	Rb	Cs
Z	3	11	19	37	55
Z_{eff}	1.3	2.5	3.5	5.0	6.4

Atomic radius *increases* down the group – due to progressive filling of new shells (main quantum number, n)

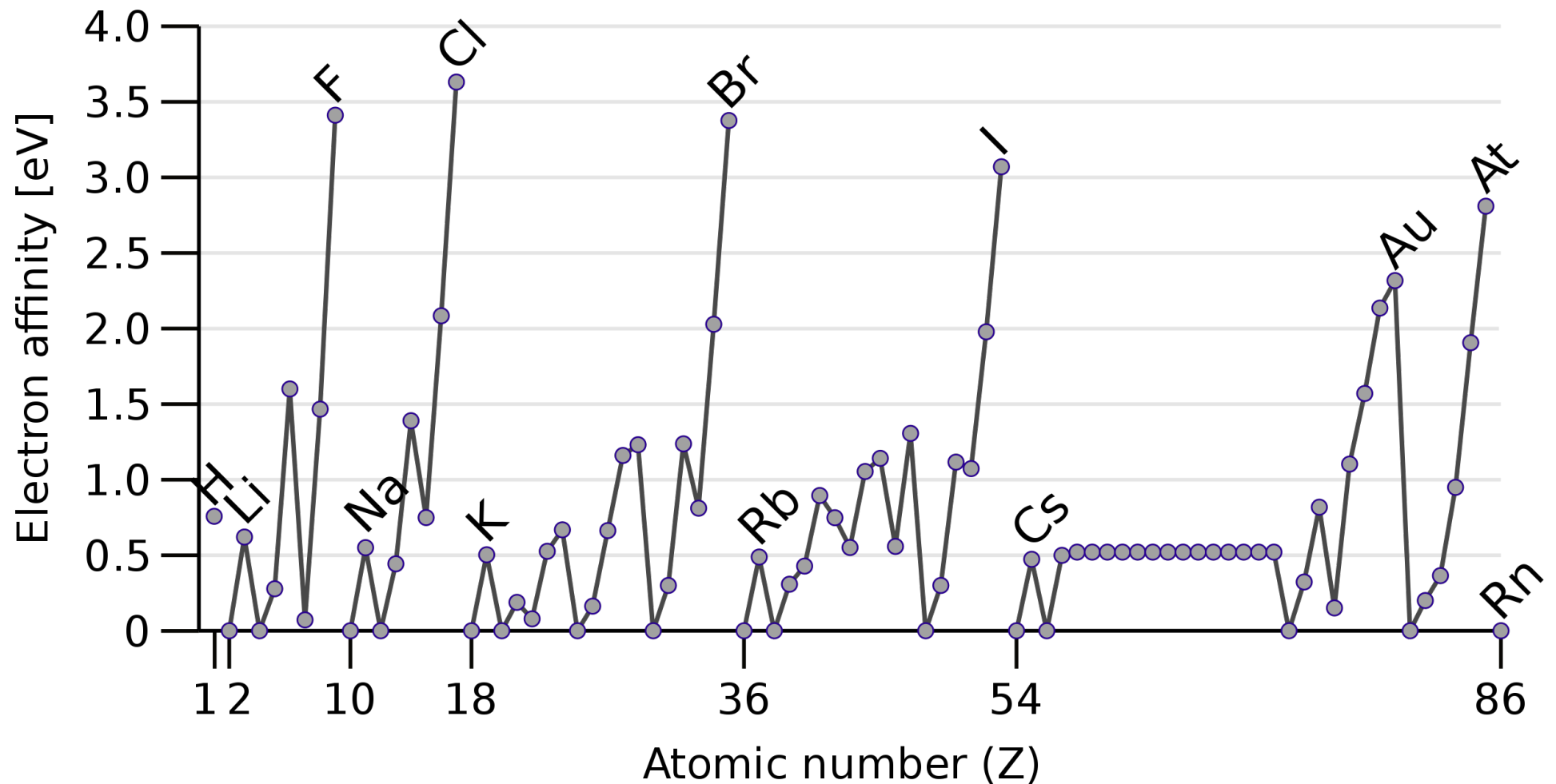
Atomic radius – along periods



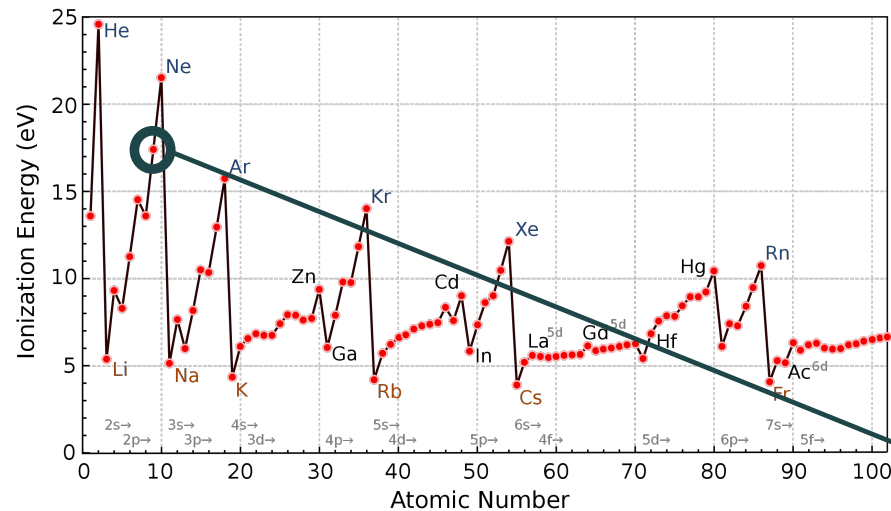
Ionization energy



Electron affinity

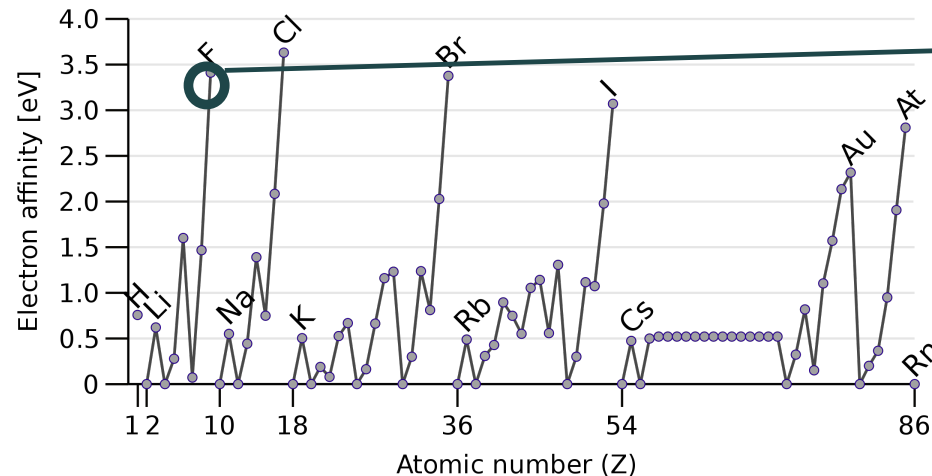


Electronegativity



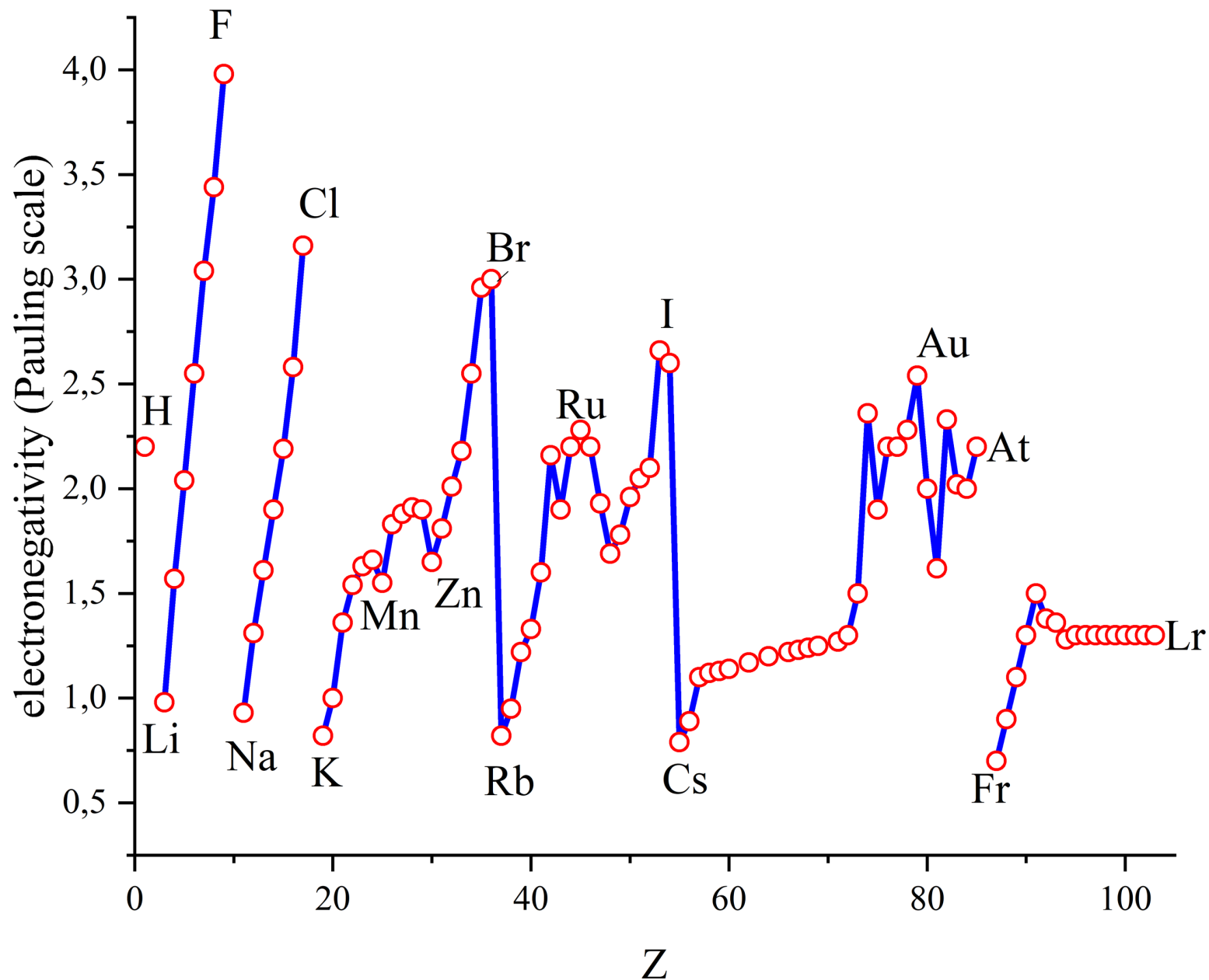
Mulliken's (simple) definition

$$\chi = (IE + EA)/2$$

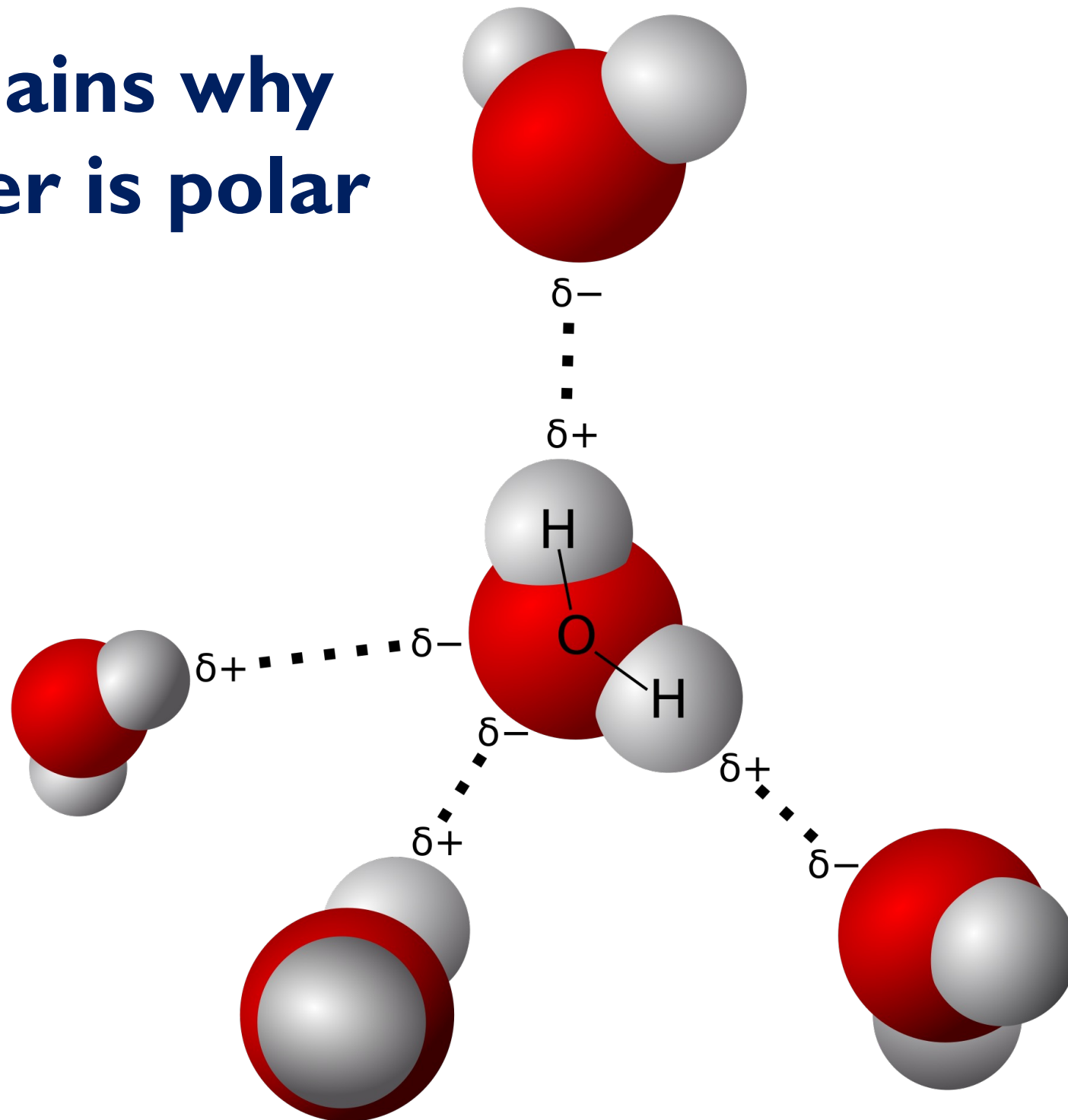


Fluorine is the most electronegative atom

Electronegativity



Explains why water is polar



Ionic radius

Sizes of atoms and their ions in pm

Group 1		Group 2		Group 13		Group 16		Group 17	
Li^+	Li	Be^{2+}	Be	B^{3+}	B	O	O^{2-}	F	F^-
90	134	59	90	41	82	73	126	71	119
Na^+	Na	Mg^{2+}	Mg	Al^{3+}	Al	S	S^{2-}	Cl	Cl^-
116	154	86	130	68	118	102	170	99	167
K^+	K	Ca^{2+}	Ca	Ga^{3+}	Ga	Se	Se^{2-}	Br	Br^-
152	196	114	174	76	126	116	184	114	182
Rb^+	Rb	Sr^{2+}	Sr	In^{3+}	In	Te	Te^{2-}	I	I^-
166	211	132	192	94	144	135	207	133	206